

Smartti Pulse

Hackathon Data Specification v1.0

This document describes the JSON data package provided to hackathon teams. The dataset contains building energy consumption and indoor climate data from three NovaProp properties, collected by the Smartti Pulse platform.

File	Property	Size	Readings	Incidents
aurora_house.json	Aurora House, Innovation Drive 14, Northville	28 MB	179,216	90
meridian_tower.json	Meridian Tower, Solar Avenue 3, Coastview	72 MB	422,983	231
horizon_plaza.json	Horizon Plaza, Central Square 26, Lakeside	29 MB	190,307	153
metadata.json	Index / summary	<1 KB	—	—

JSON Structure

Each property file follows this top-level structure:

```
{
  "metadata": { "version", "exported_at", "source" },
  "customer": { ... },
  "property": {
    "id", "name", "address", "city", "zip_code",
    "region_code", "timezone", "gross_area_m2",
    "smartti_commissioned_at", "smartti_automation_id",
    "pricing": { ... },
    "nodes": [ ... ],
    "readings": { metric: { count, earliest, latest, data: [...] } },
    "incidents": [ ... ],
    "children": [ ... ]    // Meridian Tower only
  }
}
```

Customer

Field	Type	Description
<code>id</code>	string	Unique customer identifier
<code>name</code>	string	Customer name ("NovaProp")
<code>region_code</code>	string	Country code (ISO 3166-1 alpha-2)
<code>currency</code>	string	Currency code (ISO 4217)
<code>timezone</code>	string	IANA timezone
<code>default_pricing</code>	object	Default energy prices (see Pricing)

Property

Field	Type	Description
<code>id</code>	string	Unique property identifier
<code>name</code>	string	Property display name
<code>address</code>	string	Street address
<code>city</code>	string	City
<code>zip_code</code>	string	Postal code
<code>region_code</code>	string	Country code
<code>timezone</code>	string	IANA timezone; all reading timestamps are UTC

Field	Type	Description
<code>gross_area_m2</code>	number null	Gross floor area in m ²
<code>smartti_commissioned_at</code>	string null	ISO 8601 date when Smartti was activated
<code>smartti_automation_id</code>	string null	Automation system identifier
<code>pricing</code>	object	Property-level pricing (overrides customer defaults)
<code>nodes</code>	array	Devices and systems in the property
<code>readings</code>	object	Measurement data grouped by metric
<code>incidents</code>	array	Maintenance events and fault reports
<code>children</code>	array undefined	Sub-properties (Meridian Tower only: buildings 1, 2, 3)

Pricing

Energy prices per property. Property-level values override customer defaults. If null, use the customer default.

Field	Type	Unit	Description
<code>electricity_eur_per_MWh</code>	number null	€/MWh	Electricity price
<code>district_heating_eur_per_MWh</code>	number null	€/MWh	District heating price
<code>district_cooling_eur_per_MWh</code>	number null	€/MWh	District cooling price
<code>water_eur_per_m3</code>	number null	€/m ³	Water price

Nodes

Nodes represent physical devices, systems, or buildings within a property.

Field	Type	Description
<code>id</code>	string	Unique node identifier
<code>name</code>	string	Display name (e.g., "TK01 Toimisto", "Rakennus 1")
<code>type</code>	enum	<code>device</code> <code>system</code> <code>building</code>

Field	Type	Description
tags	string	Comma-separated tags
ekohuolto_id	string	External maintenance system ID

Readings

Readings are grouped by metric. Each metric contains a summary and a data array.

```
"readings": {
  "electricity_kwh": {
    "count": 45453,
    "earliest": "2021-02-10T12:00:00+00",
    "latest": "2026-05-21T20:00:00+00",
    "data": [
      { "t": "2021-02-10T12:00:00+00", "v": 123.45, "key": "28337/1", "src": "enerkey" }
    ]
  }
}
```

Reading Data Point

Field	Type	Description
t	string	ISO 8601 timestamp (UTC)
v	number	Measured value
key	string	Sensor/meter key (identifies the physical meter)
src	enum	Data source: enerkey smartti_automation

Available Metrics

Metric	Unit	Source	Description
electricity_kwh	kWh	enerkey	Electricity consumption (hourly)
district_heating_MWh	MWh	enerkey	District heating consumption (hourly)
district_cooling_MWh	MWh	enerkey	District cooling consumption (hourly)
water_m3	m³	enerkey	Water consumption (hourly)
interior_temperature_C	°C	smartti_automation	Indoor temperature
interior_temperature_target_C	°C	smartti_automation	Temperature setpoint

Metric	Unit	Source	Description
interior_temperature_tolerance_C	°C	smartti_automation	Allowed deviation from setpoint
interior_co2_ppm	ppm	smartti_automation	Indoor CO ₂ concentration

Data Sources

Source	Description	Typical Interval
enerkey	Energy metering system (electricity, heating, cooling, water)	1 hour
smartti_automation	Building automation system (temperature, CO ₂)	~5–15 minutes

Incidents

Maintenance events, fault reports, and energy optimization actions.

Field	Type	Description
id	string	Unique incident identifier
node_id	string null	Related node (device/system)
created_at	string	ISO 8601 timestamp (UTC)
updated_at	string	ISO 8601 timestamp (UTC)
event_type	string	"Vikailmoitus", "Energiansäästötoimenpide", "Muu huomio"
priority	enum	immediate significant according_to_agreement
status	enum	pending in_progress awaiting_review resolved
category	enum	fault optimization administrative
points	number	Scoring value (positive = resolved, negative = open/stuck)
description	string	Free-text description (Finnish)
internal	boolean	If true, internal note (not shown to customer)
include_in_report	boolean	If true, included in customer reports

Field	Type	Description
<code>comments</code>	array	Thread of comments on this incident

Incident Comments

Field	Type	Description
<code>id</code>	string	Comment identifier
<code>created_at</code>	string	ISO 8601 timestamp
<code>modified_at</code>	string null	Last modification timestamp
<code>comment</code>	string	Comment text (Finnish)
<code>status</code>	string	Status at time of comment
<code>commenter_name</code>	string null	Name of commenter

Children (Meridian Tower only)

Meridian Tower has three sub-buildings as child properties. Each child has the same structure as a property (nodes, readings, incidents). The parent `meridian_tower.json` contains building-level automation data (indoor climate), while children contain per-building energy metering (electricity).

Notes for Hackathon Teams

1. **All timestamps are UTC.** Properties are in Finland (Europe/Helsinki = UTC+2 / UTC+3 DST).
2. **Energy data is hourly** (enerkey source). Indoor climate data has irregular intervals (~5–15 min).
3. **Readings key** identifies the physical meter — multiple keys per metric means multiple meters in the same property.
4. **Pricing** can be used to calculate energy costs: `reading_value × price`. Note: `electricity_kWh` readings are in kWh but price is €/MWh, so divide by 1000.
5. **Incidents are in Finnish.** Common types: "Vikailmoitus" = fault report, "Energiansäästötoimenpide" = energy saving action.
6. **Meridian Tower** has a parent-children structure. The parent contains automation data (temperatures), children contain per-building electricity.